Do Now · retrieval
Iteration in C#

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is the difference between = and == in C#?

Q2

Give one reason Unicode may be chosen instead of standard ASCII.

Q3

Convert 01011000 0011 to denary using an 8-bit two’s-complement mantissa and a 4-bit two’s-complement exponent.

Q4

mark=85. The selection is: if mark >= 80 then print("Distinction"); else if mark >= 50 then print("Pass"); else print("Fail"). Determine the output and explain which branch runs.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Iteration repeats instructions — Iteration means executing a block of instructions repeatedly. The loop needs a rule that determines whether another repetition happens.

Core idea 2

for = count-controlled — CODE:
for (int i = 0; i < 5; i++)
{
Console.WriteLine(i);
} — The header sets the counter, tests whether the loop continues, then changes the counter after each repetition.

Core idea 3

while = test before repeating — A while loop is useful when you do not know how many attempts will be needed. — CODE:
while (valid == false)
{
// ask again
}

Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Core idea 4

do...while = run first, test after — CODE:
do
{
Console.Write("Password: ");
answer = Console.ReadLine();
}
while (answer != "computer"); — The body executes before the condition is checked, so it always runs at least once.

Vocabulary · use the correct technical terms

Iteration — Repeating a block of instructions.

Count-controlled iteration — Repetition where the number of repeats is known or controlled by a counter.

Condition-controlled iteration — Repetition that continues according to a Boolean test.

Counter — A variable used to track the number of repetitions.

Independent skill check · apply the lesson

Write a C# while loop that asks for an age until it is between 0 and 120 inclusive. Test -1, 121 and 0.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Build a times-table program

Follow the worked method

1. Read the number — CODE:
Console.Write("Number: ");
int number =
int.Parse(Console.ReadLine()); — This happens once before the repetition begins.

2. Calculate each multiple — CODE:
for (int i = 1; i <= 5; i++)
{
int answer = number * i;
Console.WriteLine(answer);
} — The same two instructions run five times, but i changes each time.

3. Trace a known input — CODE:
number = 7

i = 1 → 7
i = 2 → 14
i = 3 → 21
i = 4 → 28
i = 5 → 35

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · solo guided practice

Work ALONE. Try first; read the clues one at a time if you get stuck.

Complete the steps below. Check your answer only on page 8.

YOUR TASK

Complete the pattern
CODE:

```
for (int i = ?; i <= ?; i++)
{
Console.WriteLine(?);
}
```

CLUES · uncover one at a time

Clue 1: The counter starts at 1 and increases by 1 each time.

Clue 2: The loop must include 5 but must not run when i becomes 6.

Clue 3: The value printed each time is the counter multiplied by itself.

BUILD YOUR ANSWER · in BLACK PEN

Step 1: Complete the loop start and condition.

Step 2: Complete the counter update.

Step 3: Write the output statement and test the five results.

CHECK BEFORE TURNING THE PAGE

Have you shown every step? The worked self-check is at the END on page 8.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Times-table generator

Ask for a whole number and output its first 10 multiples.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

Password attempts
Use do...while to keep asking until the password is "computer" .
Count how many attempts were made and output the total after the correct password.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A while loop always executes at least once.

Use key vocabulary from memory

1. Explain Iteration in your own words.

2. Explain Count-controlled iteration in your own words.

3. Explain Condition-controlled iteration in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

END SELF-CHECK · only after you have attempted the work

PAGE 5 · Guided We Do solution:

for (int i = 1; i <= 5; i++) { Console.WriteLine(i * i); } Output: 1, 4, 9, 16, 25.

PAGE 3 · Independent skill check:

Use do-once input before while OR initialise age to -1: while (age < 0 || age > 120) { age = int.Parse(Console.ReadLine()); }. The first two entries are rejected; 0 is accepted.

Correct any errors in GREEN PEN. Write one improvement.

YOU DO ANSWERS: rhsccs.org/lessons/h446/lesson-010/answers.html